

Customer Shopping Behavior Analysis

A personal analytics project analyzing 3,900 retail transactions to understand what actually drives repeat purchases and loyalty.

1. Why I Took This On

I wanted a project that forced me to work across the full analyst stack -- cleaning data in Python, writing real business-question SQL, and presenting findings in a dashboard a non-technical stakeholder could actually use. This dataset (3,900 transactions, 18 features spanning demographics, purchase details, and shopping behavior) had enough depth to ask meaningful questions about who buys, why they come back, and what pushes them toward a purchase.

My goal going in was to answer one core question: *what patterns in this data can a retail team actually act on to increase repeat purchases and spend its marketing budget more effectively?*

2. Dataset Summary

3,900 rows, 18 columns, covering customer demographics (age, gender, location, subscription status), purchase details (item, category, amount, season, size, color), and shopping behavior (discount usage, promo codes, previous purchases, purchase frequency, review rating, shipping type). The Review Rating column had 37 missing values I needed to handle before analysis.

3. Cleaning & EDA in Python

I started in pandas with `df.info()` and `df.describe()` to get a feel for distributions and spot anything unusual before touching the data.

Missing values: Imputed the 37 missing Review Ratings using the median rating per product category rather than a single global median or mean -- review ratings are ordinal and skew toward the 3-5 range, so a category-level median avoids flattening real differences between, say, footwear and outerwear ratings.

Column standardization: Renamed everything to snake_case early on -- a small thing, but it saved friction later when writing SQL against the same table.

Feature engineering: Binned age into age_group (Young Adult / Adult / Middle-aged / Senior) so I could compare cohorts instead of 53 individual ages. I also derived a rough purchase_frequency_days value from the categorical Frequency of Purchases field (e.g. Weekly -> 7, Every 3 Months -> 90) since the dataset has no actual transaction date -- this was the closest thing to a recency signal I could construct.

Redundant columns: Checked whether discount_applied and promo_code_used were carrying the same information. They were almost perfectly overlapping, so I dropped promo_code_used to avoid double-counting the same signal in later group-bys.

Database load: Connected the cleaned DataFrame to a local MySQL database so the rest of the analysis could be done in SQL rather than pandas -- partly to practice writing production-style queries instead of one-off notebook code.

4. SQL Analysis (MySQL)

I loaded the cleaned data into a local MySQL database and wrote queries to answer ten core business questions about customer behavior, loyalty, and purchase drivers.

1. Revenue by Gender

Compared total revenue generated by male vs. female customers.

gender	revenue
Female	75191
Male	157890

2. High-Spending Discount Users

Identified customers who used discounts but still spent above the average purchase amount.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
...	...

3. Top 5 Products by Rating

Found products with the highest average review ratings.

item_purchased	avg_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

4. Shipping Type Comparison

Compared average purchase amounts between Standard and Express shipping.

shipping_type	avg_purchase
Standard	58.46
Express	60.48

5. Subscribers vs. Non-Subscribers

Compared average spend and total revenue across subscription status.

subscription_status	total_customers	avg_spend	total_revenue
Yes	1053	59.49	62645.00
No	2847	59.87	170436.00

6. Discount-Dependent Products

Identified 5 products with the highest percentage of discounted purchases.

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. Customer Segmentation

Classified customers into New, Returning, and Loyal segments based on purchase history.

customer_segment	num_customers
Loyal	3116
Returning	701
New	83

8. Top 3 Products per Category

Listed the most purchased products within each category.

category	item_purchased	total_orders
Accessories	Jewelry	171
Clothing	Blouse	171
Footwear	Sandals	160
Outerwear	Jacket	163

9. Repeat Buyers & Subscriptions

Checked whether customers with more than 5 purchases are more likely to subscribe.

subscription_status	repeat_buyers
No	2518
Yes	958

10. Revenue by Age Group

Calculated total revenue contribution of each age group.

age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

5. Power BI Dashboard

I built the dashboard around the questions a store manager would actually ask, not just a wall of charts: total customers, average purchase amount, average review rating, revenue and sales split by category and age group, and a subscription-status breakdown, all filterable by gender, category, and shipping type.

6. Recommendations

1. Prioritize converting non-subscribers, not just retaining subscribers.

Non-subscribers generated \$170,436 (73%) of total revenue against \$62,645 from subscribers -- the bigger opportunity is turning high-spending non-subscribers into subscribers, not just rewarding the smaller group that already converted.

2. Re-engage the 'Returning' segment before they churn.

701 customers sit in Returning, a meaningful gap below the 3,116 already Loyal. This is the group most likely to move to Loyal with the right nudge (e.g. a loyalty milestone offer) -- and the most likely to slip backward if ignored.

3. Review discount depth on high-discount-dependency items before assuming it's working.

Hats, Sneakers, and Coats are discounted in ~48-50% of purchases. That level of reliance on discounting is worth questioning rather than treating as a success story -- it may be propping up demand for items that wouldn't sell as well at full price, which affects margin more than the revenue numbers alone show.

4. Target Express-shipping customers with slightly higher-value offers.

Express shipping customers spend \$60.48 on average vs. \$58.46 for Standard -- a small but consistent gap suggesting this group is a little less price-sensitive and may respond better to premium bundles than discount codes.

5. Focus loyalty incentives on the 958 repeat buyers who haven't subscribed yet.

958 non-subscribers have already made more than 5 purchases -- they behave like loyal customers without the subscription revenue attached, making them the most efficient group to target for conversion rather than running a broad campaign.

7. Limitations & What I'd Do Next

- No real transaction date exists in this dataset, so the recency proxy I built from the categorical purchase-frequency field is an approximation, not a true days-since-last-purchase measure. A real dataset with timestamps would let me build an actual RFM model instead.
- The dataset is a single snapshot, not a time series, so I can't tell whether the segment sizes (e.g. 80% 'Loyal') are stable or the result of how the categories were defined -- I'd want to sanity-check the segmentation thresholds against a second dataset if I had one.
- Given more time, I'd build a simple churn-likelihood model (logistic regression) on top of the engineered features instead of relying only on rule-based segmentation.